



XCELERate

Step up to build future of mobility!



The Pain Points of EV Industry

Raw Material

India lacks **reserves of lithium** and cobalt, so it is dependent on countries like **Japan and China** for **import** of lithium as well as lithium-ion batteries

Financial Constraints

The **initial cost** of owning an electric car is **higher** than that of ICE vehicles mainly due to the cost of the battery installed in the vehicles

Poor Infrastructure

Lack of communication system-based charging infrastructure. Every EV-making company has its own different charging port which is a hurdle to setting up a proper charging ecosystem

Range Anxiety

There is a **lack of charging facilities** in India. Despite steps being taken by the government, rural areas still lack infrastructures causing people to fear that during their travels they might not find a charging station

Common Myths

There are **existing myths** around electric vehicles including long charging duration, costlier maintenance, safety issues, high prices, battery durability, reusability and recyclability

Service Alerts

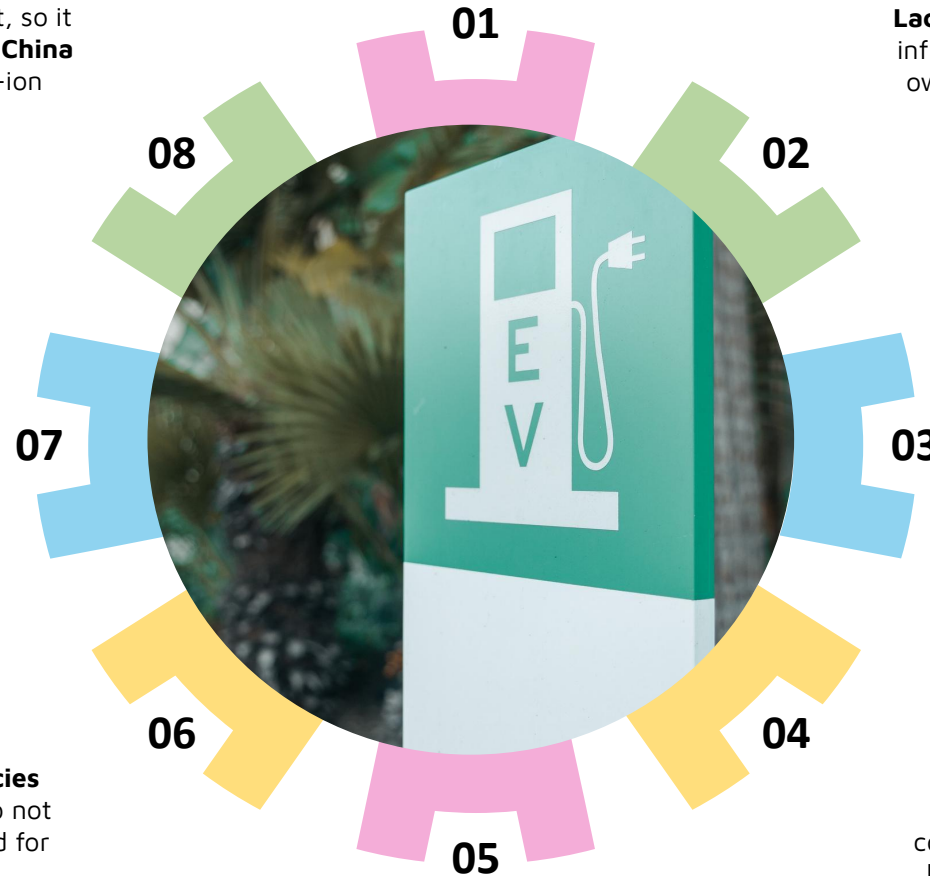
When one is stuck with their broke ICE vehicle, they can easily find **help/customer support** but when it's about EVs it's a **tedious task** to find someone who can fix their issue or help them

Temperature Issues

Temperature affects the performance of an EV battery at a large extent making EVs **inappropriate** for too cold or too hot regions where **temperature** lies **outside the range 15-40 degrees**

Government Policies

Many Indian states still **lack state policies** supporting Electric Vehicles and are also not investing to build infrastructure required for mass adoption of EVs



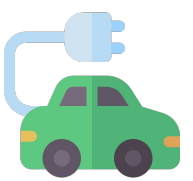
Research



Infrastructure and Facilities



Status quo of the Market



Last Mile Delivery and Car Rentals

Charging stations are present only in major **metropolitan areas** thus restricting the use of EVs by people in rural India

India's EV to Public charging station ratio is **135** as compared to the global average of **6-20**

Supply of electricity is more reliable in **urban areas** as compared to rural areas causing problems in **domestic charging** of EVs

Domestic charging also causes electricity bill to shoot up thus restricting people from owning EVs

The difference in the **Initial cost** of EV and Internal Combustion Engines (ICE) 2 wheeler compared to 4 wheeler EV and ICE is **less**

The CAGR of 2 Wheelers is **60%** as compared to **45%** for 4 Wheelers

People in **urban areas** have less range anxiety than those in **rural areas** because of lack of education and awareness in rural areas

Lack of charging facilities means people need to travel large distances to refuel the EVs

Last mile delivery services like Zomato & Swiggy that have to reduce fuel emissions are shifting to EVs to meet emission standards

The CAGR for last mile delivery services using EVs is **18-20%**

Car rental firms such as Ola and Uber are shifting towards having EVs to become sustainable and reduce large amount of carbon emissions

Car rental firms are growing at a CAGR of **40-50%** for 2 Wheelers

Analysis

Consumer awareness regarding the reduced **Total Cost of Ownership (TCO)** for EVs as compared to ICE is increasing demand



The cost of owning an EV 2 Wheeler despite a higher upfront price, **breaks even** with ICE 2 wheeler after **3 years**

Less service charges throughout the life of vehicles, **increased prices of fuel**, **increased wear and tear** of ICE Vehicles are a few reasons for the TCO of EVs to be less than ICE

Government Initiatives

Increased government initiatives like Road tax Exemption for EVs and investment of 14000 Cr to build **48,000** charging stations by 2026 has increased confidence in EVs

Government is **reducing customs duty** on import of ores required for manufacturing battery thus reducing making charges of EVs



Market Analysis



2 Wheeler Market

- **Demand:** At present it is **0.63 million** growing at a **CAGR of 60%**

Apart from private vehicles, even last mile delivery services require 2 Wheeler EVs in order to curb carbon emission problems

- **Market:** It is very **fragmented** and has various manufacturing units taking equal share in the market

The highest share is by **OLA Electric** which is **17%**, but OLA is not a conventional vehicle manufacturing firm and thus does not increase the confidence of people in using EVs

- **Benefits:** A fragmented market depicts that as such **no single company** has been able to **influence** the customers and this can act as an opportunity for Maruti to tap into and cater to the increasing demands for EV 2 Wheelers
- **Cons:** A Fragmented market means that customer has no preference so all firms will be selling similar products at a similar price hence Maruti will have to incorporate **unique features** or **reduce prices** to attain good market share



4 Wheeler Market

- **Demand:** At present it is **0.038 million** growing at a **CAGR of 45%**

There is a shift towards use of EV 4 Wheelers amongst aware citizens as well as by car rental firms

- **Market:** At present **TATA** has a **monopoly** in the EV Market, with a **market share of 80%**

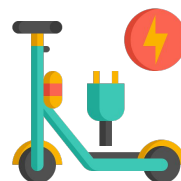
TATA Motors takes the advantage of its various subsidiaries like **TATA Metaliks** for mining raw material, **TATA Power** for developing charging stations

- **Benefits:** Customers never prefer a monopoly and Maruti has a market share of around **46% in ICE Vehicles** as compared to mere **13.9% for TATA**, depicting positive news for Maruti
- **Cons:** Being a monopoly TATA Motors might try to leverage its position & make **barrier to entry** for Maruti Suzuki very **high** & TATA also has a head start in the industry & has more than one model of EV 4 Wheelers in market

Target Customers



People living in **cities**, areas with **maximum charging stations** & belonging to **middle and upper class** household, ready to pay higher initial cost for 4 Wheelers



Families owning **2 Wheelers** willing to make transition cause of comparable **initial cost**, reduced **service charges** in the future & lower **Total Cost of Ownership**



Large businesses like last mile delivery services and car rental firms who need to meet emission standards because of pollution and are willing to make the change



Our Goal

Find the Target Audience

Take Advantage of Various Factors

Plan an Adoption Strategy

External Factors

There are certain factors in India that are going to help Maruti make their mark in the EV Industry

Large Population

India has a **large population**, leading to rise of cheap labour allowing to **educate and create skilled labourers** willing to work at fair prices

01

Government Investments

Government of India is investing money to tackle fuel emissions thus helping people buying EVs as well as car manufacturers designing EVs

02

ICE Vehicles Share

Maruti Suzuki already has made its mark in the Indian Market and has a **46% share in ICE Vehicles** hence gaining market share in electric vehicle Industry in the beginning will be easy

03

Competitor Analysis

Competitors like **TATA Motors** have only ventured in 4 Wheeler vehicles and 2 Wheeler market is very **fragmented** with equal distribution of market share amongst many firms

04

Internal Factors

TDSG Manufacturing Plant

TDSG is India's first Lithium-ion Battery manufacturing plant in Gujarat, set up jointly between **TOSHIBA, DENSO & Suzuki Motor Corporation** to manufacture & supply Lithium-ion battery. This will help reduce the cost of manufacturing

05

WagonR EV Testing

Earlier in 2019, Maruti had **tested an electric vehicle based on its Wagon R hatchback**. This road-testing will help Maruti Suzuki as it contributes in large measure to guaranteeing the quality of the entire vehicle, vehicle systems, or components.

06

2-Wheeler Manufacturing

Maruti Suzuki is also launching a **manufacturing unit for two Wheelers** in Haryana. This plant can be used for the production of electric two wheelers and cater to the large demand and capture the fragmented EV 2 Wheeler Market

07



The Beginning of an Era: Market Adoption

Elements of a perfect Adoption Strategy



Achieve the company's **Strategic Objectives**



Keeping in mind the **Constraining Factors**



Achieving the **Target** and being **Profitable**

Objectives

Partnering up with **last mile delivery services** that use 2 wheelers (3.5-4 kwh) and **car rental firms** that have to meet emission standards

Benefits

Last mile delivery firms have a huge demand for 2 Wheeler EVs which will help **increase sales**

- Zomato plans on 100% conversion to EVs by 2030
- Amazon requires 1 lakh EVs for delivery by 2030
- Flipkart plans to deploy around 25000 EVs by 2025

Car rental firms amidst push from Indian Government for pollution control are demanding EVs

- Uber has a target for 100% conversion to EVs by 2040 to curb carbon emissions in India

This will help in **building familiarity to EVs** for individual customers and reduce their fear of using it and raise sales from this sector as well

Objectives

Targeting initial manufacturing of 4 Wheeler with a battery size in the range of **20 to 25 kwh** and keeping **DC Fast charging** option

Benefits

Battery size of mid range should be chosen first as more the battery range more expensive the product

- EV 4 Wheelers do not have comparable initial cost with ICE counterparts thus a mid range battery helps in **reduction of initial price**
- A 25-30 kwh battery range in a 4 wheeler will help it easily cover a distance of 250 kms in a single charge

Higher battery range Electric Vehicles can cover more distance as compared to mid range ones after a single charge

- In future when Maruti starts becoming profitable more models need to be **introduced with larger battery range**

EV should have the **DC Fast charging** option as that will help in using a Level 3 charger which **reduces time for charging** the vehicle

Objectives

Leverage relation with **Toshiba and Denso** to create **Lithium ion batteries** to reduce initial cost of EV and export batteries in the long run

Benefits

In house manufacturing of Lithium ion batteries will help **reduce manufacturing cost** of making an EV

- Currently Lithium ion batteries are imported
- Importing batteries increases price of manufacturing EV ultimately increasing initial cost of EV
- It will help gain **subsidy from Government** under "**Make in India**" programme

A proper research centre for making Electric Vehicles will help **improve the battery efficiency** as well

- A more efficient battery helps in covering **larger distance** in a single charge which can **reduce range anxiety**

This can also **increase revenues** by selling batteries to other EV Sellers and **exporting batteries** to other countries as well



The 3 Cs to Cater

Collaboration for Growth

- Taking advantage of **Maruti Udyog and Suzuki collaboration** and partnering up with government for **standardisation of charging ports** of EVs to further increase sales
- A type 2 charger is compatible for Level 2 and Level 3 AC and Level 3 DC Charging. Level 3 DC Charging helps in reducing the charging time as well, hence more focus should be on such charging stations
- Chargers compatible to Level 1 stations should also be provided for emergency usage in old stations and for home charging

Communication for Engagement

- **Partnering with schools** and promoting benefits of EVs and marketing the reduced **Total Cost of ownership of EVs** e.g. According to Mckinsey Report the TCO of EV 2 Wheeler is **\$2.48/100 Km** compared to **\$3.78/100 Km** for ICE at the end of 7 years to increase awareness in younger minds

Creation for Improvement

- Venturing into **battery switching to reduce charging time** after establishing battery making facility and becoming profitable
- Establishing Battery Switching stations will help **reduce** one of the greatest bottlenecks of using Electric Vehicle that is **charging time** (battery switching at present is only possible for 2 wheelers)
- Maruti has also allocated some resources for creating a recycling centre in Gujarat, which can be made into EV Recycling facility to reuse the components of Lithium Batteries

Points to Consider



- EVs need to be designed to heat or cool off the battery in order for the battery to perform at its best in the extreme temperatures of India



- Upskilling workers is a must as servicing and regular check of EVs require specialised skills and training



- Customers will be provided with training regarding working and charging of EV, along with a hotline number for any disruptions while travelling

Demand Forecasting

A Standard 3 Years (2023-2025) is required for the full setup

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Overall Demand for 2 Wheelers (Mn)	0.63	0.945	1.417	2.1262	3.1893	4.7840	7.1760	10.7641	16.1462	24.2193	36.3289	54.4934
Demand for Maruti (Mn)				0.1701	0.3189	0.5740	1.0046	1.7222	2.9063	4.8438	7.9923	13.0784
Overall Demand for 4 Wheelers (Mn)	0.038	0.055	0.079	0.1158	0.1679	0.2435	0.3531	0.5121	0.7425	1.0767	1.5612	2.2637
Demand for Maruti (Mn)				0.0115	0.0184	0.0292	0.0459	0.0716	0.1113	0.1722	0.2654	0.4074

According to a Mckinsey Report EV 2 Wheelers in India have a current demand of **0.63 Mn** growing at **CAGR of 60%**

In a fragmented market, where all firms at present have market share ranging from 12-17%, an initial **market share of 8%** for Maruti is considered with a steady **2% increase** every year

According to a Mckinsey report EV 4 Wheelers in India have current demand of **0.038 Mn** growing at **CAGR of 45%**

Assuming **10% market share** for 4 Wheelers in a market dominated by TATA, with steady **1% increase**

Supply Capabilities

Maruti is opening a new manufacturing unit in **IMT Kharkhoda, Haryana**, which is **9 times** the current **100 acre** facility in Gurugram

The manufacturing unit will be used specifically for manufacturing **EVs** and is the biggest venture of Suzuki Motorcycle India since its starting in 2006

There are plans to manufacture **0.06 Million 4 Wheeler Vehicle** and **0.15 Million 2 Wheeler Vehicle** in the initial year

800 acres will be utilised for manufacturing **4 wheelers** and **100 acres** by Suzuki Motorcycles for manufacturing **2 Wheelers**

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
2 Wheeler supply by Maruti (Mn)				0.15	0.165	0.1815	0.1996	0.2196	0.2415	0.2657	0.2923	0.3215
4 Wheeler supply by Maruti (Mn)				0.06	0.060	0.0612	0.0618	0.0624	0.0630	0.0636	0.0643	0.0649

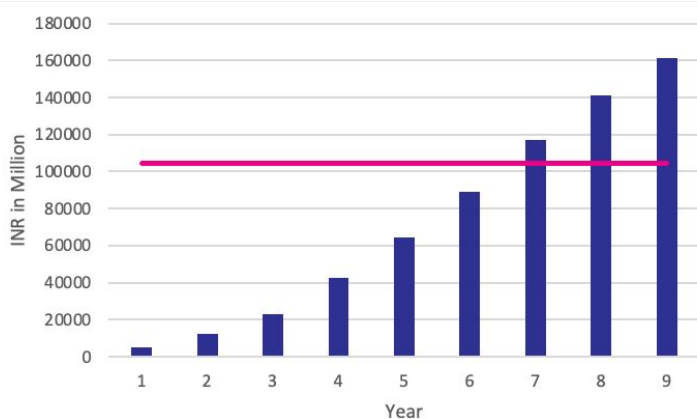
The growth rate of **2 Wheelers** is **10%** and **1%** for **4 Wheelers** as, the manufacturing of 4 Wheelers is more difficult and takes longer duration as compared to 2 Wheelers

In the future Maruti will also venture into supplying **different model of EVs** eg. 4 Wheeler with higher battery range to improve the kms run/one time charge

Supply of Batteries

TDSG, India's first **Lithium-ion Battery** manufacturing plant in **Gujarat** set up jointly between **TOSHIBA, DENSO & Suzuki Motor Corporation** will start by **2026** for manufacturing batteries required by Electric Vehicles

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
Revenue for 2 Wheelers (Mn)				22500	24750	27225	29947.5	32942.25	36236.47	39860.12	43846.13	48230.74
Revenue for 4 Wheelers (Mn)				11584.7	18477.7	29228.38	61818.06	62436.24	63060.6	63691.21	64328.12	64971.4
Total Revenue (Mn INR)				34084.7	43227.7	56453.38	91765.56	95378.49	99297.07	103551.3	108174.2	113202.1
Operating profit (Mn INR)				5112.71	7348.71	10726.14	19270.76	21937.05	24824.26	27958.85	23798.33	20376.38
Net Profit (Mn INR)				-99287.2	-91938.5	-81212.4	-61941.6	-40004.6	-15180.33	12778.5	36576.8	56953.2



From the table and graph we can depict that Maruti will be able to break even after 6 years of starting its operation

The profit will **increase more than anticipated** because of added revenues from the possibilities of **export of battery** produced in the TDSG Facility in Gujarat

Avg price of 2 wheelers = 1.5 lakhs
Avg price of 4 wheelers = 10 lakhs
Revenue = Minimum (Supply,Demand) * Price

Operating profit of EV manufacturers are 10-15%. Assuming **15% operating profit**, Maruti will be **increasing** it at a **steady 2%** due to inhouse battery manufacturing and other reduction in expenses

Initial Investment

Total Investment (Mn INR)	104,400
Investment for Vehicle Manufacturing (Mn INR)	31000
Investment for Battery Manufacturing (Mn INR)	73000

Net Profit = Cumulative Operating Profit - Investment

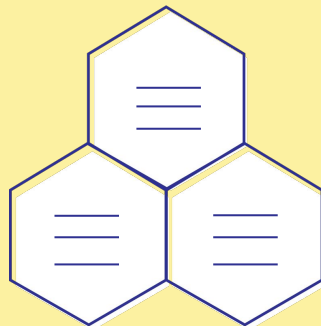
Striving for Improvement

2 Wheeler Production: For the 2 Wheeler Market the demand is exponentially greater than the supply
The revenue can be increased by increasing production capacities for 2 Wheeler and meeting demand of the last mile delivery firms as well as individual customers

Government Subsidy:

The government is favouring the transition of ICE Vehicles to EV so Maruti can look into subsidy programmes

The TDSG Plant manufacturing batteries caters to the "Made in India" concept, hence getting further support from Government will be easier



Numerous Models:

As the revenue keeps on increasing with a decrease in expenses, Maruti should build more models

Higher battery range vehicles fetch more price in the market and will also reduce the range anxiety for people who can afford high end products

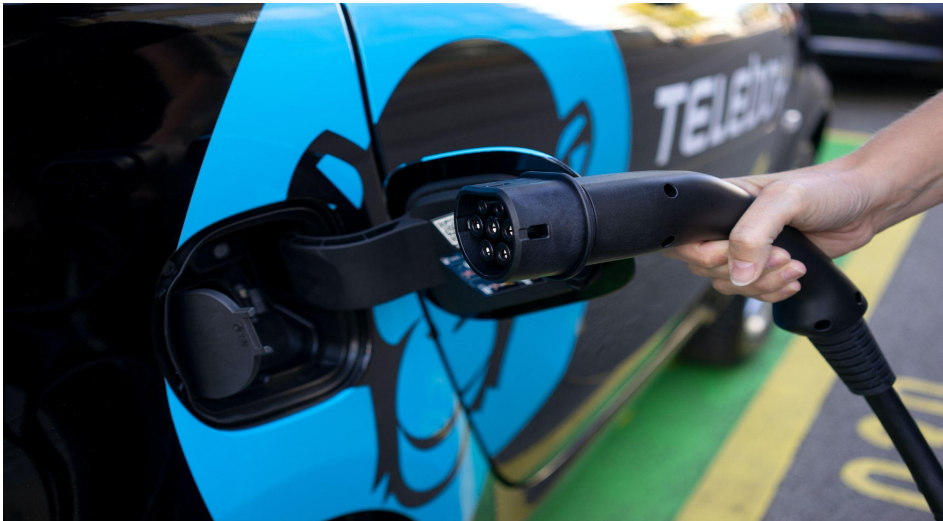


Solar EV Charging

Issue: Cost of Charging an EV at home is determined by the respective state's electricity rate. Charging EV at a proper **charging station** costs around **Rs 100-200**, and **home charging costs** around **Rs 180-500**. In rural areas there is **lack** of constant **supply of electricity** for **Domestic Charging**

Solution: Collaboration with government for charging stations with solar rooftop photovoltaic facility

- An EV with solar is **51% cheaper** than charging on grid power and **80% cheaper** than charging on public chargers
- EV in **USA** cause **greater reduction of CO2 emissions than in India** since the grid energy (used by the EV) in India is predominantly generated from coal and from US has charging stations with solar rooftop photovoltaic facility



Hydrogen Fuel Cell Electric Vehicles

Issue: In around than **10 years**, it will become **cheaper to run a fuel cell electric vehicle (FCEV)** than a battery electric vehicle (BEV) or an internal combustion engine (ICE) commercial vehicle, according to a report published jointly by Ballard and Deloitte China.

Solution: Maruti Suzuki in future should venture into Hydrogen based fuel vehicles

- The central government recently said it is considering green hydrogen to fuel vehicles. India had also recently launched its National Hydrogen Energy Mission (NHEM) to promote the use of the natural gas.
- FCEVs are currently more expensive to run per 100km than BEVs and ICE vehicles, but through proper R&D, infrastructure development, maturing manufacturing technologies they can become cheaper and TCO can be reduced to half

Jack of all Trades

Year 2040: Maruti ventures into battery switching to reduce charging time of batteries, keeps on increasing battery efficiency with the help of the facility developed in partnership with Toshiba and Denso and launches other model of EVs

Partners in Growth

Year 2032: Maruti collaborates with government to build standardise charging station to increase user comfort, partners up with schools for better awareness by targeting young minds

Baby Steps

Year 2025: Maruti launches first EV 2 Wheeler and 4 Wheeler in the market with production of own Lithium-ion Batteries in just the next year and the power starts shifting from countries exporting Lithium-ion batteries to India

Breaking Finances

Year 2031: Sales from customers as well as business like last mile delivery services and car rentals keep on increasing and Maruti breaks even and achieves a huge Milestone. Revenue also increase from the export of Lithium-ion Batteries

Maruti Suzuki and the EV Empire

Year 2050: Maruti becomes a name worldwide in the EV domain, exporting not just vehicles but even raw materials like lithium batteries and gaining a huge Market Share in both 2 and 4 Wheeler Sector and at the same time venturing into newer technologies like hydrogen fuel vehicles



Thank You!